



Lab Worksheet 8

CSE461: Introduction to Robotics

Department of Computer Science and Engineering

Lab 08: Scenario-Based Problem Solving: From Concept to Prototype

I. Topic Overview

In this lab, students will use the components and skills they learned throughout the semester to solve small real-life style problems. Students will choose the appropriate sensors, actuators, and controllers, design the system, and implement the logic themselves. By completing this lab, students will demonstrate their ability to integrate inputs, outputs, and control strategies to build a functional, self-designed robotic solution.

II. Learning Outcome

After this lab, students will be able to:

1. Identify the appropriate sensors, actuators, and controllers needed to solve a given scenario.
2. Design and build simple circuits that combine multiple input and output devices.
3. Test, debug, and troubleshoot both hardware connections and program logic to achieve a working solution.

Each group will only solve a single scenario selected by the faculty

IV. Scenario A : Smart Pet Feeder with Day/Night Control

- **Objective:** Build a small automated feeder system that dispenses food only when the environment is bright AND a trigger button is pressed.
- **Available Components:**
 - Arduino, Raspberry PI
 - LDR
 - Push Button
 - Servo motor
 - LED
 - LCD display for live status
- **Requirements:**
 1. Light Detection: Use the LDR to determine whether it is daytime.
 - If brightness > threshold → allow feeding
 - If brightness < threshold → feeding disabled
 2. User Trigger: When the button is pressed AND it is daytime, activate the servo to rotate 90° to "dispense food" for 1–2 seconds. And the servo will go back to its original position.
 3. Status Indication:
 - LED ON when feeding is allowed (daytime).
 - LED OFF when it is night or too dark.
 4. On the LCD display, show the below texts during the respective actions:
 - Feeding: Allowed / Not allowed
 - “Dispensing...” when the servo moves

V. Scenario B : Environment-Aware Ventilation System with Manual Override

- **Objective:** Create a small ventilation system that automatically reacts to temperature/light conditions and allows a manual override.
- **Available Components:**
 - Arduino, Raspberry PI
 - LDR
 - Push Button
 - DC motor
 - LED
 - LCD display for live status
- **Requirements:**
 1. Automatic Mode (default): The system decides when to turn ON/OFF the “ventilation fan” (DC motor) based on light conditions
 - If the LDR detects bright light → run the DC motor
 - If light is very low → turn the motor OFF
 2. Manual Override Mode: The button toggles between:
 - Auto Mode
 - Manual Mode

In Manual Mode: The motor automatically turns ON, ignoring the LDR readings. If the button is pressed again, the mode reverts back to Auto mode and the motor is controlled by the LDR value.
 3. Status Indication:
 - LED ON when the motor is running.
 - LED OFF when the motor is off.
 4. On the LCD display, show the below texts during the respective actions:
 - Mode: Auto / Manual (Based on the mode)

VI. Deliverables:

- Circuit diagram
- Arduino code
- Demonstration of the working circuit